

Roll No.

TME-304

B. TECH. (ME) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022

BASIC THERMODYNAMICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

(iii) Draw neat diagrams as necessary.

(iv) Make assumptions for any missing data.

1. (a) State and explain the first law of thermodynamics for a closed system.

(CO1)

OR

(b) What are the different types of processes for a closed system ? Explain the zeroth law of thermodynamics.

(CO1)

2. (a) Derive the expression for work done in a polytropic process.

(CO1)

OR

(b) One cubic meter of an ideal gas at 25 deg C is compressed isentropically to one tenth its volume. Find the resulting pressure and work done. Take the isentropic index of compression to be 1.4 and initial pressure to be 100 kPa.

(CO1)

P. T. O.

(2)

3. (a) Explain the concept of compressibility factor with the help of a chart.

(CO1)

OR

- (b) Explain the concept of dryness fraction. Find the specific internal energy, specific enthalpy and specific entropy of steam at a pressure of 10 bar and a dryness of 0.68.

(CO1)

4. (a) Derive the steady flow energy equation.

(CO2)

OR

- (b) Air at 100 kPa and 300 K is compressed steadily to 660 kPa and 450 K. The mass flow rate is 0.1 kg/s and heat loss of 16 kJ/s occurs during the process. Find the power input to the compressor and the specific work done.

(CO2)

5. (a) Explain the form of the first law of thermodynamics when applied to turbines and compressors along with the units of all the quantities involved.

(CO2)

OR

- (b) The gas turbine of a turbojet engine receives a steady flow of high pressure gases at 10 bar and 900 deg C at a velocity of 150 m/s. It discharges the gases at a pressure of 1.15 bar, a temperature of 450 deg C and a velocity of 200 m/s. Determine the specific external work output. Assume the constant pressure specific heat to be 1.004 kJ/kg-K.

(CO2)